

MAUVE-MUSE: A Resolved View of Star Formation, Gas Cycling, and Chemical Enrichment in Virgo Cluster Galaxies

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Format: 45-minute talk followed by approximately 15 minutes of Q&A

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Purpose and narrative

The talk begins with the baryon cycle as the general framework for galaxy evolution: gas enters a galaxy, changes phase, forms stars, becomes enriched, and can return through feedback and recycling. PHANGS then provides the resolved observational benchmark, showing how nearby star-forming discs can be connected across molecular gas, H II regions, feedback, chemical enrichment, gas and stellar dynamics, and stellar populations at roughly 100 pc scales. PHANGS is not a purely isolated control sample, and the displayed NGC4254 field is itself in Virgo; the point is that PHANGS established the resolved method. The environmental question then becomes explicit. Much of the classic cluster picture comes from integrated or coarse-resolution measurements: cold gas is removed outside-in, total star formation tends to fall, and global gas-phase metallicity can be enhanced at fixed stellar mass. Those trends do not reveal which link in the baryon cycle changed inside the surviving disc. Virgo's proximity and environmental information make it the ideal laboratory for that resolved experiment, leading directly to MAUVE. The talk then uses the resolved metallicity-SFR relation as the scientific centrepiece. Its gas-regulator interpretation is developed as a scale-free analytic competition between SFR or SFE variability and gas-supply variability, capable of producing either sign from 100 pc regions to whole galaxies. The final section connects this framework to ongoing work on where star formation remains during ram-pressure stripping, preliminary evidence that spiral structure shapes metallicity fluctuations in NGC4254, and molecular-gas tests.

The order follows the latest Yonsei discussion in [Seminar Package Advice](#). The accepted broad title is retained. The KIAS title and narrower emphasis are not used here. The timing is 45 minutes of speaking; questions are additional.

Central message: MAUVE connects environmental transformation to the balance of gas supply, star formation and chemical enrichment. The observed metallicity-SFR coupling depends on local stellar surface density and metallicity diagnostic. One scale-free regulator equation explains how SFR/SFE-driven variability can produce a positive correlation while supply-driven variability produces a negative one, at both resolved and integrated scales. Preliminary NGC4254 results show that spiral geometry, enrichment memory and clustered star formation can leave distinct signatures in metallicity fluctuations.

Timing at a glance

Section	Slides	Duration	Clock	Role in the argument
Motivation: baryon cycle, PHANGS benchmark and environmental gap	1-4	5:00	00:00-05:00	Move from the general cycle to resolved field-disc methods, then ask what changes inside cluster galaxies
MAUVE and selected collaboration results	5-9	6:00	05:00-11:00	Show why a resolved multiphase survey is needed
Paper I: metallicity and star formation	10-17	13:00	11:00-24:00	Present the evidence, spatial behaviour and diagnostic limits
Gas regulator: one analytic competition across scales	18-25	8:30	24:00-32:30	Derive both signs, map resolved to integrated variables, and position the contribution
Ongoing work: where star formation remains	26-31	7:30	32:30-40:00	Distinguish SF occupancy, non-detection and detected non-SF emission
Metallicity fluctuations and molecular-gas tests	32-35	4:00	40:00-44:00	Present the preliminary NGC4254 decomposition and define the reservoir test
Conclusions and discussion	36	1:00	44:00-45:00	Reconnect environment, gas cycling and enrichment

The main presentation contains 36 slides. Six additional slides support questions and are outside the 45-minute budget. Slides are paced by the argument and by figure explanation, rather than by a fixed one-slide-per-minute rule.

1. Motivation: 00:00-05:00

Slide 1. Title and introduction - 0:30

Use the accepted title, presenter, affiliation and Yonsei date, with an existing MUSE galaxy image. Acknowledge Luca Cortese, Barbara Catinella, Luke Davies and the MAUVE collaboration verbally and in the notes. The opening is about the wider survey and physical problem, rather than only the paper title.

Slide 2. Galaxy evolution is regulated by the baryon cycle - 1:30

Begin with the general physical picture. Galaxies grow because gas enters from the halo and circumgalactic medium, joins the interstellar medium, cools into denser phases, and forms stars. Stellar evolution returns both gas and newly synthesised metals, while feedback drives outflows that may escape, remain in the circumgalactic medium, or recycle. The rates need not balance instantaneously, but their long-term competition regulates gas content, star formation and chemical enrichment. Use the baryon-cycle artwork as a conceptual map, not as a quantitative measurement. This establishes the language used later in the resolved gas-regulator section. [S11]

Transition: That picture is usually introduced for galaxies evolving in relative isolation. A cluster adds external forces and can change several links in the cycle at the same time.

Slide 3. PHANGS maps the baryon cycle at ~100 pc - 1:30

Introduce the observational step from a conceptual cycle to linked local measurements. PHANGS combines MUSE spectroscopy with ALMA molecular-gas maps and complementary HST/JWST imaging to connect molecular clouds, H II regions, current star formation, feedback, chemical enrichment and mixing, gas and stellar kinematics, and stellar populations in nearby star-forming discs. Use the local NGC4254 MUSE VRI composite as the visual anchor. State the sample qualification explicitly: PHANGS is a resolved nearby-disc benchmark rather than a purely isolated field control, and NGC4254 itself is a Virgo member. The key achievement is the method and physical scale, not an environment-free sample definition. [S13]

Transition: PHANGS shows that the baryon cycle can be resolved. What changes when cluster environment becomes the organizing axis of the sample?

Slide 4. What does a cluster change inside the disc? - 1:30

Start with the established integrated picture: cluster satellites undergo outside-in cold-gas removal, often have reduced total SFR, and can show higher global gas-phase metallicity at fixed stellar mass. Say that much of this classic evidence is integrated or coarse-resolution, rather than implying that no resolved cluster studies exist. Integrated measurements establish that environment matters but cannot distinguish continued stripping, reduced replenishment, compression and H I-to-H₂ conversion, altered molecular star-formation efficiency, or feedback inside the surviving disc. Existing resolved studies already hint that inner H I and H₂ can differ from isolated-galaxy relations, but available data do not discriminate among the competing mechanisms. Virgo at approximately 16.5 Mpc offers roughly 100-200 pc physical resolution, full gas-disc coverage and well-developed infall-stage information. Frame the resolved question as: which link changes, where, and when? [S12]

Transition: MAUVE was designed to perform that resolved environmental experiment across the Virgo infall sequence.

2. MAUVE and selected collaboration results: 05:00-11:00

Slide 5. MAUVE resolves the Virgo environment - 1:15

Expand the acronym: Multiphase Astrophysics to Unveil the Virgo Environment. Describe the 40-galaxy survey design across infall stages, MUSE ionized gas and stellar spectroscopy at roughly 100-200 pc scales, ALMA molecular gas, and complementary HST stellar/cluster information. Existing atomic-gas observations add wider environmental context. Distinguish programme design from the available subset used in an individual analysis. [S1, S12]

Slide 6. MAUVE early science connects feedback, gas phases and ionization - 1:15

Follow the presentation structure used by Luca Cortese and Toby Brown: introduce the early-science projects as one connected set before discussing individual systems. Use three visual cards. NGC4383 follows an outflow across ionized, molecular and atomic phases and reaches a fountain interpretation. NGC4064 asks whether feedback can complete quenching after environmental stripping. The 12-galaxy Brown et al. comparison asks why non-HII emission becomes more visible as star formation fades. The common message is that the cluster changes the physical setting in which feedback, recycling and ionization operate; MAUVE science is broader than gas removal alone. [S2-S5, S16, S17]

Slide 7. NGC4383: from ionized wind to a multiphase fountain - 1:00

Use the four-step evidence chain foregrounded in both talks. A central starburst launches a roughly 6 kpc ionized bicone. Molecular gas contributes little above approximately 1 kpc. Atomic hydrogen likely dominates the cold component farther from the disc. Cold-gas velocities remain below those of the ionized phase and below escape, favouring bound circulation in a galactic fountain. Retain the main limitation: the warped H I disc prevents a precise decomposition and cold mass-loading estimate. [S2, S3, S16, S17]

Slide 8. NGC4064: feedback may complete quenching after stripping - 1:00

Adopt the physical framing used by Luca and Toby. NGC4064 is a gas-poor backplash system whose outer reservoir has already been strongly reduced, but concentrated central star formation drives an ionized bipolar outflow extending about 2 kpc. Present the paper's estimates: ionized outflow rate 0.53 solar masses per year, integrated SFR 0.32 solar masses per year, and mass loading $\eta = 1.66$. Under the assumed density, geometry and filling factor, the inferred removal timescale is 0.58 Gyr compared with 0.97 Gyr for consumption by star formation. Conclude that feedback may accelerate final quenching in this object, with the stated systematic uncertainties. [S4, S16, S17]

Slide 9. As HII emission fades, diffuse ionized gas becomes visible - 1:30

Make Brown et al.'s causal ordering clearer. The study compares 12 early MAUVE-MUSE galaxies with 12 matched PHANGS-MUSE field galaxies at roughly 100 pc resolution. Environmental quenching suppresses bright HII emission, allowing pervasive diffuse ionized gas to become the dominant visible ionized component. Turbulence within selected star-forming regions remains similar between samples. A small high-dispersion tail above 100 km/s, containing 5% of MAUVE spaxels versus 1% in PHANGS, supports a secondary contribution from shocks or turbulent mixing layers. The BPT figure on the slide shows the ionization shift; the velocity-dispersion evidence comes from Figure 11. [S5, S16, S17]

Transition: These results make it essential to select the SF-dominated emission carefully before interpreting metallicity and SFR together.

3. Paper I: metallicity and star formation: 11:00-24:00

Slide 10. The local link between stars, star formation and metals - 1:00

Define stellar mass surface density, SFR surface density and gas-phase oxygen abundance. Briefly distinguish the resolved SF main sequence, the resolved mass-metallicity relation and the debated secondary SFR dependence. Use the EAS conceptual context figure, with its original attribution. [S6, S7]

Slide 11. Selecting star-forming emission - 1:30

Show the original BPT diagnostic figure and state the paper's HII selection: secure HII classification in both BPT diagrams, $EW(H\alpha) > 6$ Angstrom, and intrinsic H α dispersion < 45 km/s. SFR is derived from dust-corrected H α . The analysis uses 14 galaxies from internal release v2; ensemble relations exclude metallicity outlier NGC4383 and therefore contain 13 galaxies. Do not import the incorrect dispersion inequality printed in the old EAS slide. [S6]

Slide 12. The resolved mass-metallicity relation - 1:30

Show the original O3N2-M13 and O3N2-C20 relation panels. Explain the primary increase and high-density flattening before adding the SFR dependence. Identify stellar surface-density units explicitly. The two O3N2 recipes share an index and are not independent spectral diagnostics. [S6, Figure 2]

Slide 13. At fixed stellar density, the SFR dependence changes - 2:00

Use the original SFR-stratified relation and explain the low-density and high-density regimes separately. Higher local SFR is associated with lower O3N2-inferred abundance at low stellar surface density and with higher abundance at high surface density. Describe a conditional association, not an observed temporal response. [S6, Figure 3]

Slide 14. The sign changes near log stellar surface density 7.5-8.0 - 2:00

Show the Spearman-versus-density figure with its original galaxy colours and the black ensemble reference. Define the sign of the coefficient. State the approximate transition in $\log_{10}(\Sigma_{\text{star}} / [\text{solar masses kpc}^{-2}])$; it is not an exact universal threshold. This figure supplies the quantitative version of the preceding visual result. [S6, Figure 5]

Slide 15. Opposite local behaviours coexist in one disc - 1:30

Use the original NGC4192 offset and Moran-like map. Explain that correlated and anti-correlated spatial patterns can occur within one galaxy. The map is a local association diagnostic; neither its colour nor its absolute index is by itself a causal measurement or a calibrated significance map. [S6, Figure 4]

Slide 16. Metallicity calibration changes the inferred relation - 2:00

Compare the four diagnostics in an editable summary: O3N2-M13 and O3N2-C20 show the coherent inversion; D16 is mostly mildly positive across much of the density range; S-calibration PG16 is more heterogeneous and can show the opposite ordering. The primary rMZR is more stable than the secondary relation. Discuss ionization, N/O, DIG and shared line measurements as relevant systematics. Keep this caveat in the main talk. [S6, Section 5.1 and Figure 6]

Slide 17. What the observations establish - 1:30

Summarize the O3N2-dependent sign reversal and the coexistence of local behaviours. Separate the observed association from the question of its physical cause. Environment may modulate the gas cycle, but the Paper I result alone does not establish ram-pressure stripping as the cause of the inversion. [S6]

Transition: A useful model should explain how both signs can arise locally and identify the additional quantity we need to measure.

4. Gas regulator: one analytic competition across scales: 24:00-32:30

Slide 18. A local molecular reservoir - 0:45

Use the existing EAS reservoir illustration, clearly as a schematic. Define the local molecular gas reservoir, its gas supply, star formation and removal. Local supply means replenishment from the surrounding ISM, not automatically pristine accretion from outside the galaxy. The reservoir boundary matters. [S6, S7, S10]

Slide 19. Gas continuity - 1:15

Introduce the gas budget, with all rates per unit area:

$$\Sigma_{\text{SFR}} = \frac{\Sigma_{\text{g}}}{\tau_{\text{dep}}} ,$$

$$\frac{d\Sigma_{\text{g}}}{dt} = \Sigma_{\phi} - (1 - R)\Sigma_{\text{SFR}} - \Sigma_{\text{out}} .$$

Walk through supply, long-lived stellar locking and removal. State that instantaneous recycling and a single effective reservoir are model assumptions. Removal can include feedback and environmental effects, but the equation does not measure either rate from the MUSE maps. [S6, Section 4]

Slide 20. Metal enrichment competes with dilution - 1:15

Use the abundance equation:

$$\frac{dZ}{dt} = \frac{y(1 - R)\Sigma_{\text{SFR}} + (Z_{\phi} - Z)\Sigma_{\phi}}{\Sigma_g}.$$

Explain the positive enrichment term and the dilution term when supplied gas has lower abundance. Here Z is a linear abundance, not the plotted $12 + \log_{10}(\text{O}/\text{H})$. The explicit removal term cancels only under the assumption that removed gas has the reservoir abundance; removal still changes the reservoir, SFR and response time. [S6, Section 4]

Slide 21. The instantaneous target abundance - 1:15

Introduce the target and relaxation time:

$$Z^{\dagger} = Z_{\phi} + y(1 - R)\frac{\Sigma_{\text{SFR}}}{\Sigma_{\phi}},$$

$$\tau_{\phi} = \frac{\Sigma_g}{\Sigma_{\phi}}, \quad \dot{Z} = \frac{Z^{\dagger} - Z}{\tau_{\phi}}.$$

At fixed yield, recycling and supplied abundance, the first-order relation is

$$\delta Z^{\dagger} = (Z^{\dagger} - Z_{\phi})$$

$$\times (\delta \ln \Sigma_{\text{SFR}} - \delta \ln \Sigma_{\phi}).$$

Natural logarithms are used here. Supply rising proportionally more than SFR lowers the target; SFR rising proportionally more than supply raises it. This is a statement about the target under stated assumptions, not an unconditional theorem about the sign of an observed correlation. [S6]

Slide 22. The response has a finite timescale - 1:00

Explain that the observed abundance lags and smooths a changing target. A fixed-target response is used only as an analytic illustration. Correlations depend on how the drivers covary, their timescales and the phases sampled. A spatial map does not directly measure a time derivative. Do not identify the effective relaxation time uniquely with a measured turbulent-mixing time. [S6, Section 4.2]

Slide 23. One equation produces both correlation signs - 1:00

Present the analytic result as a competition between relative driver amplitudes. If SFR or SFE fluctuations dominate over supply fluctuations, enrichment produces a positive metallicity-SFR correlation. If gas-supply fluctuations dominate while supply and SFR co-vary, dilution produces a negative correlation. Comparable amplitudes yield a weak relation. Express the same statement as δZ proportional to $\delta \ln \tau_{\Phi}$ minus $\delta \ln \tau_{\text{dep}}$. The finite response time adds phase lag and scatter without changing the identities of the two competing drivers. [S6, Sections 4.1-4.3]

Slide 24. The regulator is scale-free - 1:00

Make the resolved-to-integrated mapping explicit. Replace the surface-density variables (Σ_g , Σ_* , Σ_{SFR} , Σ_{Φ}) with their galaxy-integrated counterparts (M_g , M_* , SFR, Φ). The continuity equations and perturbation rule retain the same functional form. At fixed local stellar density or fixed total stellar mass, SFR/SFE-driven variability can therefore produce a positive correlation, while gas-supply-driven variability can produce a negative one. Emphasize that scale-free algebra does not require identical timescales, covariance or spatial averaging at the two scales. [S6, Section 5.3]

Slide 25. What this adds to earlier regulator models - 1:00

Position the result by physical mechanism. Many bathtub applications foreground stochastic inflow and therefore recover the familiar negative FMR. Forbes et al. (2014) primarily model that accretion-driven anticorrelation and discuss how scatter in parameters, including mass loading, can change the covariance; this feedback-sensitive route is not the preferred mechanism here. Wang and Lilly (2021) explicitly demonstrate the positive SFE-driven route and the negative inflow-driven route using small-amplitude analytic approximations and numerical response experiments. The contribution of this work is a compact analytic competition between relative SFR/SFE variability and gas-supply variability, written once and carried from resolved to integrated scales. Do not say that Wang and Lilly provided only numerical solutions: their paper contains approximate analytic expressions that they verify numerically. [S6; Forbes et al. 2014; Wang and Lilly 2021]

5. Ongoing work: where star formation remains: 32:30-40:00**Slide 26. Where star formation remains along the RPS sequence - 1:00**

Introduce the ongoing occupancy question. Present the analysis snapshot as 26 product-level systems after inclination and product-availability selection: 8 pre-peak, 5 close-to-peak and 13 post-peak. NGC4567_8 is one combined product representing two physical components. These counts belong to this analysis, not the complete 40-target survey. [S8, S9]

Slide 27. SF, ND and NSF measure different outcomes - 1:30

Define SF as the selected HII/EW/dispersion population; ND as failure of the required Balmer detections; NSF as Balmer-detected emission that fails the SF selection. Define the denominator as the valid observed footprint. ND is sensitivity-limited, and NSF may mix DIG, shocks, old-star and nuclear excitation. The categories do not uniquely identify physical mechanisms. [S8]

Slide 28. SF occupancy is lower in the post-peak sample - 2:00

Use editable charts made directly from the existing equal-galaxy summary CSVs. Show the full valid footprint alongside the post-fit $S/N \geq 25$ subset. Full-footprint mean SF fractions are 0.388, 0.251 and 0.066 for pre-, close- and post-peak; corresponding ND fractions are 0.312, 0.509 and 0.727. Mark the result as preliminary and cross-sectional. Do not interpret the S/N cut as a neutral removal of bad data: it changes the denominator and preferentially excludes ND area. [S8]

Slide 29. Outer non-detections and central non-SF emission - 1:15

Show the existing equal-galaxy radial profiles, with cut and uncut curves explicitly identified. The descriptive picture combines a contracting SF footprint with a broader decline in occupancy; the destination differs with radius. Outer and low-density regions are increasingly ND, while detected NSF emission becomes prominent centrally. Avoid treating the stage average as the track of every galaxy. [S8, S9]

Slide 30. The maps show substantial galaxy-to-galaxy diversity - 1:15

Use selected existing spatial maps, including contrasting close-to-peak systems and a post-peak example. Keep SF/ND/NSF/excluded colours and labels unchanged. Connect this diversity to the small close-to-peak sample and the need for galaxy-level inference. Do not label an individual NSF feature as a confirmed shock or AGN using the category map alone. [S8, S9]

Slide 31. Occupancy and surviving-SF intensity are separate - 0:30

State the next measurement: compare the SFR of regions that remain SF at fixed stellar density and radius. The present fractions cannot tell whether those survivors are enhanced, normal or suppressed. [S8]

6. Metallicity fluctuations and molecular gas: 40:00-44:00**Slide 32. NGC4254: spiral structure shapes metallicity fluctuations - 1:00**

Present the user-provided four-panel NGC4254 snapshot as a preliminary result. The upper-left panel shows the dereddened H-alpha field and fitted source spirals; the upper-right panel shows the metallicity fluctuation used in the correlation. The lower-left panel compares the measured two-point correlation with the uniform KT18 baseline and spiral-aware model. The lower-right panel decomposes the complete correlation. The result demonstrates that spiral geometry leaves a measurable imprint on the enrichment pattern in this galaxy. Keep the scope explicit: this is preliminary evidence from NGC4254, not a sample-wide measurement. [S15]

Slide 33. Four terms separate geometry, memory and source clustering - 1:00

Define each correlation component precisely. $C_{\{2p,0\}}$ is the uniform-field injection-diffusion prediction of Krumholz and Ting (2018). $C_{\{2p,N\}}$ adds the perturbed spiral field. Its azimuthal integral is zero for complete, symmetric arms, but asymmetric arms or incomplete observational masks can leave a nonzero net term. C_M is the mean or cumulative past correlation over the correlation time, allowing a visible spiral imprint to remain in the fluctuation phase. C_{NP} is the first-order residual from clustered sources after injection-diffusion: one GMC forms correlated clusters whose Gaussian diffusion couples stellar drift to the enrichment yield delay. This born-in source correlation is physically distinct from both the injection-diffusion baseline and the cumulative mean. [S15]

Slide 34. ALMA measures the reservoir and depletion time - 1:30

Define Σ_{H_2} , $\tau_{\text{dep}} = \Sigma_{\text{H}_2} / \Sigma_{\text{SFR}}$ and molecular SFE. Explain the test of whether the positive branch associates with depletion-time changes or whether gas abundance and supply proxies are more informative. CO constrains molecular mass after conversion assumptions; it is not a direct map of the local replenishment rate. Match beams, apertures, sensitivity and tracer timescales. [S1, S10]

Slide 35. A connected research programme - 0:30

Connect the three ongoing directions: environmental SF occupancy, internal metallicity structure, and molecular-reservoir tests. Emission-line products and simulation comparisons support diagnostic robustness and physical interpretation; avoid making them a separate long project list. [S0]

7. Conclusions: 44:00-45:00

Slide 36. Three messages to take into the discussion - 1:00

1. MAUVE resolves how gas, star formation and enrichment coexist within environmentally processed Virgo galaxies.
2. The main O3N2-based result is a stellar-surface-density-dependent reversal of the metallicity-SFR association; its diagnostic dependence is part of the result.
3. Environmental occupancy, the preliminary NGC4254 fluctuation decomposition and molecular gas provide complementary tests of the regulator interpretation.

Close with the question of which local conditions determine whether replenishment or enrichment dominates. Acknowledge the collaboration and invite discussion.

Detailed speaking narrative

This section is a rehearsal script for the 34-slide main talk. It is intentionally more detailed than the visible slide text. The wording can be adapted during delivery, but the order, quantitative statements and scientific qualifications should be retained.

Narrative for slide 1. Title and introduction

Good afternoon, and thank you for the invitation. I will present work from the MAUVE collaboration on how the Virgo environment changes star formation, gas cycling and chemical enrichment inside galaxies. I would especially like to acknowledge Luca Cortese, Barbara Catinella, Luke Davies and the wider MAUVE team. The central idea is that a galaxy does not respond to environment as a single number. Gas can disappear from some parts of a disc, remain in others, change phase, form stars, or become enriched. MAUVE lets us separate these outcomes spatially. I will first introduce the physical motivation and survey, then focus on our resolved metallicity-SFR result, and finish with ongoing tests using environmental stage, internal structure and molecular gas.

Narrative for slide 2. Galaxy evolution is regulated by the baryon cycle

I want to begin with the broad framework. A galaxy grows through a baryon cycle. Gas enters from the halo and circumgalactic medium, joins the interstellar medium and, after cooling into denser phases, becomes available for star formation. Stars lock up part of that material and return another part through winds and supernovae. Those processes also create and redistribute metals. Feedback can push gas out of the star-forming reservoir; some gas may escape, while some remains in the halo and later returns. Galaxy evolution therefore reflects the balance and timing of inflow, phase conversion, star formation, enrichment, removal and recycling. The diagram is a conceptual map. Later I will write a local version of this balance as continuity equations.

Narrative for slide 3. PHANGS maps the baryon cycle at ~100 pc

The baryon-cycle diagram is useful, but observation has to turn each arrow into a measurement. PHANGS has pushed that experiment to the scales of molecular clouds and H II-region complexes in nearby star-forming discs. With MUSE we can map current star formation, ionized-gas conditions and kinematics, gas-phase abundance, and the ages, chemistry and motions of the stellar population. ALMA adds the molecular reservoir, while HST and JWST connect young stars, clusters and dust to the gas. This is why roughly 100 pc matters: instead of comparing only whole galaxies, we can compare clouds, H II regions, spiral arms, bars and centres within them. The NGC4254 image makes that resolved structure visible. I should qualify the comparison carefully: PHANGS is not a purely isolated control survey, and NGC4254 itself is a Virgo member. PHANGS supplies the resolved nearby-disc benchmark and methodology. The next step is to make cluster environment, rather than internal disc structure alone, the organizing variable.

Narrative for slide 4. What does a cluster change inside the disc?

We already know from integrated and coarse-resolution studies that cluster environment changes galaxies. Cold gas is commonly removed from the outside inward. Total star formation is often reduced, and satellites can show enhanced global gas-phase metallicity at fixed stellar mass. These are essential results, but a total SFR or one metallicity value cannot tell us which physical link changed. Did stripping continue into the inner disc? Did loss of atomic gas reduce replenishment of the molecular phase? Did pressure compress the gas and accelerate H I-to-H₂ conversion? Did the efficiency of turning molecular gas into stars change, or did feedback redistribute what remained? Resolved studies provide hints that the inner H I and H₂ reservoirs already differ from isolated-galaxy relations, but the proposed mechanisms point in different directions. Virgo is the ideal laboratory because its distance of about 16.5 megaparsecs gives roughly 100-200 pc resolution, while extensive H I, CO and dynamical information lets us cover full gas discs and organize systems by infall stage. The question is therefore spatial and temporal: which link changes, where, and when?

Narrative for slide 5. MAUVE resolves the Virgo environment

MAUVE was designed to bring that resolved experiment to the Virgo environmental sequence. The programme targets 40 late-type Virgo galaxies that retain molecular gas and span different stages of infall. MUSE supplies spatially resolved stellar populations, ionized-gas diagnostics, current star formation and gas and stellar kinematics at roughly 100-200 pc scales. ALMA and VERTICO trace the molecular reservoir, while existing H I data show the extended atomic component and stripping morphology. HST adds information on stellar structure and clusters. The row shown here demonstrates the multiphase logic for one galaxy: we can connect local gas supply, phase conversion, star formation, enrichment and dynamics rather than infer them from a single integrated measurement. It is also important to separate the programme design from individual analyses: not every paper uses all 40 targets, because data availability and quality cuts differ.

Narrative for slide 6. MAUVE early science connects feedback, gas phases and ionization

Luca Cortese and Toby Brown both introduce the early MAUVE papers as a connected portfolio before going into the individual objects. I will use the same structure. NGC4383 follows a spectacular outflow across ionized, molecular and atomic gas and asks whether that material escapes or recycles. NGC4064 asks whether feedback can finish the quenching of a system whose outer reservoir has already been stripped. The 12-galaxy ionization study asks why non-HII emission becomes more prominent as the HII component fades. These are not three unrelated examples. Together they show that the cluster changes the setting in which stellar feedback, phase conversion, recycling and ionization operate. MAUVE therefore probes much more than the morphology of stripped tails.

Narrative for slide 7. NGC4383: from ionized wind to a multiphase fountain

The NGC4383 story works best as an evidence chain. A central starburst launches a roughly 6 kpc ionized bicone. The follow-up then asks where the cold mass is. ALMA shows that molecular gas contributes little above about 1 kpc. MeerKAT reveals atomic structure aligned with the outflow, and the available mass estimates indicate that H I likely dominates the cold component at larger height. The cold gas moves more slowly than the ionized phase and remains below the escape speed, so the combined evidence favours a bound fountain that redistributes gas through the halo. The important limitation is the extended warped H I disc, which overlaps the outflow in projection and prevents a precise cold mass-loading measurement. The physical result is therefore recycling rather than a securely measured escaping cold wind.

Narrative for slide 8. NGC4064: feedback may complete quenching after stripping

NGC4064 provides the complementary late-stage case emphasized in both reference talks. Environmental processing has already removed much of the outer gas, yet concentrated central star formation drives a bipolar ionized outflow extending approximately 2 kpc. The inferred ionized outflow rate is 0.53 solar masses per year, compared with an integrated SFR of 0.32 solar masses per year, giving a mass-loading factor of 1.66. Under the adopted gas-density, geometry and filling-factor assumptions, removing the remaining gas through the outflow takes about 0.58 Gyr, compared with 0.97 Gyr for star formation alone. This supports the idea that feedback may accelerate the final stage of quenching after stripping. These estimates have substantial systematic uncertainty, so the slide presents a plausible object-level mechanism rather than a universal satellite-quenching timescale.

Narrative for slide 9. As HII emission fades, diffuse ionized gas becomes visible

Brown and collaborators compare 12 early MAUVE-MUSE galaxies with 12 matched PHANGS-MUSE field galaxies at roughly 100 pc resolution. The BPT distributions show that a larger fraction of the cluster emission lies outside the classical star-forming locus. The interpretation is driven by the changing balance of ionizing sources. Environmental quenching suppresses bright HII-region emission, allowing pervasive diffuse ionized gas to dominate the visible ionized component. The selected star-forming regions themselves have similar velocity-dispersion distributions in the two environments. A small additional tail remains: 5 percent of MAUVE spaxels exceed 100 kilometres per second, compared with 1 percent in PHANGS, and these are concentrated in non-star-forming regions. That tail is consistent with a secondary contribution from shocks or turbulent mixing layers. This ordering is more precise than attributing every elevated line ratio directly to ram-pressure shocks.

Narrative for slide 10. The local link between stars, star formation and metals

I will now turn to the central result. At each resolved location we measure stellar mass surface density, star-formation-rate surface density and gas-phase oxygen abundance. Two established relations organize these quantities. The resolved star-forming main sequence links local SFR to local stellar density, and the resolved mass-metallicity relation links abundance to stellar density. The open question is whether SFR adds a secondary dependence at fixed stellar density. Some surveys find an anticorrelation, others find a weak or absent relation, and the answer can depend on spatial scale and metallicity diagnostic. MAUVE reaches much smaller physical scales than the kiloparsec-scale context shown here, so we can ask whether the relation changes within individual discs.

Narrative for slide 11. Selecting star-forming emission

Before measuring that relation, we select regions where nebular emission is dominated by star formation. We require secure H II classification in both the nitrogen and sulphur BPT diagrams. We then require H α equivalent width greater than 6 Angstrom and intrinsic H α velocity dispersion below 45 kilometres per second. The last inequality is important: the old EAS slide contained the sign in the wrong direction, while the paper and this talk use the correct upper limit. SFR comes from dust-corrected H α . The paper contains 14 galaxies from internal release v2. NGC4383 behaves as a metallicity outlier and is excluded from the ensemble relation, leaving 13 galaxies in that pooled comparison.

Narrative for slide 12. The resolved mass-metallicity relation

The first-order result is the resolved mass-metallicity relation itself. The horizontal axis is local stellar mass surface density in solar masses per square kiloparsec, and the vertical axis is oxygen abundance. Abundance rises rapidly at low stellar density and then flattens toward the high-density end. The two panels use the O3N2 index with the Marino et al. 2013 and Curti et al. 2020 calibrations. They produce different numerical abundance scales but share the same underlying line ratio, so they should not be counted as independent spectral diagnostics. This primary relation provides the baseline. The next step asks whether regions above or below the local SFR trend sit systematically above or below the abundance relation.

Narrative for slide 13. At fixed stellar density, the SFR dependence changes

These panels reveal that the secondary dependence is not monotonic across the disc. On the left, the abundance relation is separated into SFR bins. At lower stellar surface density, regions with higher SFR tend to have lower O3N2-inferred abundance. On the right, where SFR is plotted within bins of stellar density, the cool low-density tracks slope downward. At higher stellar density, the warm tracks instead show a positive association: regions with higher SFR also tend to have higher inferred abundance. The key wording is conditional association. We observe different regions at one epoch. The plot does not show an individual region evolving through time and it does not by itself identify which physical driver changed first.

Narrative for slide 14. The sign changes near log stellar surface density 7.5-8.0

Here we quantify the visual impression using Spearman rank correlations within bins of stellar surface density. Negative coefficients mean that higher SFR accompanies lower abundance. Positive coefficients mean that the quantities rise together. The individual galaxy curves preserve the adopted galaxy colours, and the black curve represents the ensemble. For both O3N2 calibrations, the typical coefficient moves from negative at low stellar density to positive at higher density. The transition lies roughly around log stellar surface density 7.5 to 8.0 in solar masses per square kiloparsec. I will treat that as an approximate regime change, not a universal threshold, because individual galaxies differ and the value depends on sampling and diagnostic choices.

Narrative for slide 15. Opposite local behaviours coexist in one disc

NGC4192 demonstrates that the two regimes can coexist within a single galaxy. One panel shows the local SFR offset, and the spatial-association panel combines the standardized SFR offset with the locally lagged abundance offset. Positive and negative regions occupy different parts of the same disc. This is useful because it argues against explaining the entire trend only through differences between galaxies. At the same time, the map must be read carefully. The colours describe a local association diagnostic. They do not constitute a calibrated significance map, and they do not prove that one process caused the other. The result motivates a local physical model, but the map alone does not select that model.

Narrative for slide 16. Metallicity calibration changes the inferred relation

The strongest caveat is that the secondary relation depends on how metallicity is inferred. O3N2-M13 and O3N2-C20 show a coherent inversion, but they share the same line index. The D16 diagnostic is mostly mildly positive across much of the stellar-density range. The PG16 S calibration is more heterogeneous and can show an ordering opposite to O3N2. The primary mass-metallicity relation is considerably more stable than this secondary dependence. That sensitivity is scientifically meaningful because the diagnostics respond differently to ionization parameter, nitrogen-to-oxygen abundance, diffuse emission and the particular emission lines shared with the SFR measurement. I therefore present the inversion as an O3N2-based observational result with explicit diagnostic dependence.

Narrative for slide 17. What the observations establish

The observational conclusion has three layers. First, within the O3N2 framework, the conditional metallicity-SFR association changes sign with local stellar surface density. Second, positive and negative local behaviours coexist within individual discs, so the ensemble trend has spatial structure. Third, the detailed secondary relation is less robust across abundance diagnostics than the primary resolved mass-metallicity relation. These statements describe the data. The physical explanation remains open. Environment may alter gas supply, compression, removal and recycling, but Paper I alone does not demonstrate that ram pressure causes the inversion. We now need a model that can accommodate both signs and that points to measurements capable of discriminating among the possible drivers.

Narrative for slide 18. A local molecular reservoir

To interpret the two signs, I will return to the baryon cycle and make it local. Imagine a boundary around a patch of star-forming molecular gas. Material can enter that reservoir from the surrounding interstellar medium, form stars, return through stellar mass loss, or leave the chosen boundary. The exact definition matters. Local supply can include transport through the disc or conversion from another gas phase. It should not automatically be identified with pristine cosmological accretion. Likewise, removal may represent feedback, environmental displacement or transport into another phase. This deliberately simple reservoir does not capture every process. Its purpose is to make the competing terms explicit and show which quantities MUSE measures and which require additional data.

Narrative for slide 19. Gas continuity

The first equation links SFR surface density to gas surface density through an effective depletion time. The second is the local gas budget. Supply adds gas at a surface-density rate Σ_{Phi} . Star formation permanently locks the fraction one minus R into long-lived stars, while R represents rapid recycling. Σ_{out} removes gas from the selected reservoir. Every term is expressed per unit area. A single effective reservoir and instantaneous recycling are assumptions, and the equation does not tell us whether removal comes from feedback or environment. Most importantly, a static MUSE map does not measure these time derivatives or the local supply rate directly. The equation organizes the interpretation; it does not turn observed maps into a measured gas-flow history.

Narrative for slide 20. Metal enrichment competes with dilution

The abundance equation adds chemistry to the gas budget. Newly formed stars contribute metals through the yield term, while supplied gas changes the reservoir abundance through the difference between the supply abundance and the current abundance. If incoming gas has lower metallicity, that term dilutes the reservoir. Here Z is a linear abundance, so it must not be confused with the plotted quantity $12 + \log \text{oxygen over hydrogen}$. The explicit outflow term cancels only when the material leaving the reservoir has the same abundance as the reservoir itself. Even then, removal still changes the gas mass, SFR and response time. Metal-enhanced winds or selective stripping would require an additional term and could change the interpretation.

Narrative for slide 21. The instantaneous target abundance

The same equation can be rewritten as relaxation toward an instantaneous target abundance, Z_{dagger} . That target equals the supplied abundance plus an enrichment contribution proportional to the ratio of SFR to supply. The response time τ_{Φ} is the gas surface density divided by the supply rate. If yield, recycling and supplied abundance remain fixed, a small fractional increase in supply relative to SFR lowers the target, while a larger fractional increase in SFR raises it. The logarithms in the perturbation equation are natural logarithms. This result explains how negative and positive associations can both arise in the model, but it is conditional. It concerns changes in the target under specified assumptions, not an unconditional prediction for the correlation measured in a spatial snapshot.

Narrative for slide 22. The response has a finite timescale

The reservoir does not jump immediately to its target abundance. The purple curve shows the exact normalized response only for a fixed target and a constant τ_{Φ} . After one response time, about 63 percent of the adjustment has occurred. Real galaxies have targets that change as supply and star formation vary, so the measured abundance lags and smooths the driver history. The sign of a spatial correlation can therefore depend on how supply and SFR covary, how rapidly they change, and which phase of that variation we observe. A map at one time cannot measure a time derivative. τ_{Φ} is also an effective response time in this model, not automatically a direct measurement of turbulent mixing.

Narrative for slide 23. One equation produces both correlation signs

The main analytic result is that positive and negative metallicity-SFR correlations are two outcomes of the same equation. When fractional SFR or SFE variations dominate over fractional changes in gas supply, the enrichment term moves metallicity in the same direction as SFR. When gas-supply variations dominate while supply and SFR rise together, dilution moves metallicity in the opposite direction. If the two amplitudes are comparable, the residual relation is weak. The equivalent timescale form compares changes in τ_{Φ} with changes in τ_{dep} . The finite response time matters because it smooths the driver and changes the observed phase, but the physical competition remains SFR or SFE variability against supply variability.

Narrative for slide 24. The regulator is scale-free

Nothing in these continuity equations requires the reservoir to be a 100 pc patch. If I replace gas, stellar, SFR and supply surface densities by total gas mass, total stellar mass, total SFR and total inflow rate, the algebra is unchanged. The same perturbation rule can therefore be read locally at fixed stellar surface density or globally at fixed stellar mass. SFR or efficiency variability can produce a positive branch at either scale, while supply variability can produce a negative branch at either scale. The word scale-free refers to the functional form. It does not mean that a galaxy-integrated measurement has the same response time, covariance or spatial averaging as an individual region.

Narrative for slide 25. What this adds to earlier regulator models

The negative FMR is familiar because many bathtub models emphasize stochastic gas accretion: more supply raises SFR and dilutes the gas. Forbes and collaborators developed that route in detail. They also discuss how scatter in regulator parameters, including mass loading, can change the covariance and allow a positive direction, but that route depends strongly on feedback assumptions and is not the mechanism I use here. Wang and Lilly then explicitly showed that time-variable SFE can drive SFR and metallicity together while time-variable inflow drives them oppositely. Their paper combines small-amplitude analytic approximations with numerical response experiments. The advance here is to write the competition compactly in one analytic target-abundance equation and then carry the same rule from resolved regions to integrated galaxies.

Narrative for slide 26. Where star formation remains along the RPS sequence

The first ongoing project asks a simpler question than the metallicity analysis: where does selected star formation remain as the environmental stage changes? The September snapshot contains 26 product-level systems after the inclination and product-availability cuts: 8 pre-peak, 5 close-to-peak and 13 post-peak. NGC4567 and NGC4568 share one combined product, so product counts and physical galaxy counts are not identical. The comparison is cross-sectional. We observe different galaxies at different inferred stages rather than follow the same objects through time. The close-to-peak group is especially small and heterogeneous. The result is therefore a preliminary description of how the spatial occupancy of star-forming, non-detected and other detected emission changes across the sample.

Narrative for slide 27. SF, ND and NSF measure different outcomes

The category definitions determine what this analysis can say. SF regions pass the H II, equivalent-width and velocity-dispersion selection used for the star-forming population. ND regions fail at least one required Balmer-line detection. NSF regions have the Balmer detections but fail the SF selection. The denominator is the valid observed footprint with finite stellar mass, accepted spatial binning and masks, and a valid radius. ND does not mean zero star formation; it includes regions below the detection criteria. NSF also does not correspond to one mechanism. It can include diffuse ionized gas, shocks, ionization from old stars or nuclear activity. A post-fit S/N cut changes the denominator, so the cut and uncut results answer different conditional questions.

Narrative for slide 28. SF occupancy is lower in the post-peak sample

These charts give each galaxy equal weight before taking the stage mean. In the full valid footprint, the mean SF fraction declines from about 39 percent in the pre-peak group to 25 percent near peak and 7 percent after peak. Over the same ordering, ND rises from about 31 to 51 to 73 percent. The right panel applies a post-fit S/N threshold of at least 25. It shows a higher SF fraction in every group because the cut removes much of the low-S/N area that would otherwise enter ND. The cut is therefore not a neutral cleaning step. The broad conclusion is a lower SF occupancy in the post-peak sample, while the exact fractions remain conditional on the footprint definition and preliminary sample.

Narrative for slide 29. Outer non-detections and central non-SF emission

The radial profiles show where the changing categories occur. Each panel corresponds to an environmental stage, with radius expressed in effective-radius units. Dashed curves use the full valid footprint, while solid curves use the S/N-selected denominator. The outer and lower-density parts of the discs increasingly fall into ND, which is consistent with a contracting detectable star-forming footprint but still includes sensitivity effects. In the inner regions, detected NSF emission can become important, especially when classical H II emission occupies a smaller fraction of the footprint. The pattern therefore involves both spatial contraction and a broader change in occupancy. These are stage averages with equal galaxy weighting; they should not be interpreted as the deterministic radial evolution of every galaxy.

Narrative for slide 30. The maps show substantial galaxy-to-galaxy diversity

The individual maps make the diversity visible. NGC4501 and NGC4424 are both assigned to the close-to-peak group, yet their spatial category patterns differ strongly. IC3392 gives a post-peak example with central selected SF and extensive surrounding ND or NSF structure. Blue marks SF, orange marks ND, red marks NSF and grey marks regions excluded by the post-fit S/N requirement. The black contour is one effective radius. These maps explain why the galaxy, rather than each pixel, must be treated as the independent statistical unit. They also warn against assigning a physical mechanism to an individual red patch. The category map alone cannot distinguish shocks, diffuse gas, old-star ionization or nuclear emission.

Narrative for slide 31. Occupancy and surviving-SF intensity are separate

The fraction analysis measures how much of the valid footprint remains in the selected star-forming category. It does not measure whether the surviving regions form stars more strongly or weakly than comparable regions in other galaxies. A galaxy could lose most of its star-forming area while the remaining central regions are enhanced, normal or suppressed. Those outcomes imply different physical mechanisms. The next step is therefore to compare the SFR surface density of the surviving SF population at matched stellar surface density and radius, while retaining galaxy-level weighting and environmental-stage uncertainty. This distinction between occupancy and intensity is important throughout the talk: fewer star-forming regions and weaker star formation within those regions are separate observational claims.

Narrative for slide 32. NGC4254: spiral structure shapes metallicity fluctuations

This figure is the current preliminary NGC4254 result. The upper-left map shows the dereddened H-alpha field with the fitted source spirals, while the upper-right map is the metallicity fluctuation used for the correlation. The lower-left panel compares the measured PHANGS-MUSE two-point correlation with the uniform Krumholz and Ting baseline and the spiral-aware version. The lower-right panel separates the complete correlation into physical terms. The key result is that the spiral geometry is not only a visual overlay: it changes the correlation structure of enrichment. This is a one-galaxy preliminary demonstration, so I will not generalize the amplitude or component balance to the full sample.

Narrative for slide 33. Four terms separate geometry, memory and source clustering

The four terms separate effects that would otherwise be blended. $C_{2-p,0}$ is the uniform-field injection-diffusion correlation from the Krumholz and Ting 2018 framework. $C_{2-p,N}$ is the perturbation caused by the spiral source field. For complete symmetric arms its azimuthal integral cancels, but real asymmetric arms and incomplete masks can leave a net contribution. C_M carries the mean or cumulative past correlation over the relevant correlation time, so the fluctuation phase can retain a spiral imprint after the instantaneous source pattern has evolved. C_{NP} is different: it is the first-order residual from the fact that sources are born correlated. A single GMC forms related stellar clusters; Gaussian diffusion then couples their drift to the delay before metals are returned. It is therefore a clustered-birth term, rather than another injection-diffusion or cumulative-mean term.

Narrative for slide 34. ALMA measures the reservoir and depletion time

ALMA supplies the missing view of the molecular reservoir. After adopting a CO-to-H₂ conversion, we can measure molecular-gas surface density and combine it with matched SFR to obtain depletion time or molecular star-formation efficiency. We can then test whether the positive metallicity-SFR branch coincides with changes in depletion time, or whether gas content and other supply-sensitive proxies provide more information. CO alone does not measure the replenishment rate into the reservoir. The comparison must match beams, apertures, sensitivity and tracer timescales, and it must propagate uncertainty in the CO conversion factor. This turns the regulator interpretation into an observational test while preserving the distinction between the amount of molecular gas and the rate at which it is supplied.

Narrative for slide 35. A connected research programme

These projects form one connected programme. Environmental occupancy shows where selected star formation remains as the gas disc is transformed. The preliminary NGC4254 decomposition shows that spiral geometry, temporal memory and clustered star formation can shape where enrichment fluctuations appear. Molecular gas measures the reservoir and depletion time needed to test the regulator interpretation. The same emission-line products, classification choices and physical-resolution matching support all three. Forward modelling provides the common comparison layer. Together, these measurements move from a descriptive environmental sequence toward a physical account of which part of the baryon cycle changes in each part of a galaxy.

Narrative for slide 36. Three messages to take into the discussion

I will finish with three messages. First, MAUVE resolves the surviving inner discs of environmentally processed Virgo galaxies, where cold gas, star formation and chemical enrichment continue to interact after the outer reservoir has been disturbed. Second, one scale-free regulator competition can produce positive or negative metallicity-SFR correlations at both local and integrated scales, with the observed O3N2 inversion and its diagnostic dependence providing the resolved target. Third, the occupancy analysis, preliminary NGC4254 fluctuation decomposition and molecular-gas measurements provide complementary tests. The question I would like to leave for discussion is which physical driver dominates in each region, and how spiral structure and environment change that balance.

Backup slides and likely questions

Backup	Subject	Use during questions
B1 / 37	Survey and analysis sample bookkeeping	40 survey targets, 14 Paper I galaxies, 13 in the ensemble relation, 26 product systems in the ongoing occupancy snapshot
B2 / 38	Full metallicity-diagnostic comparison	Original four-diagnostic figure; calibration-dependent sign and scatter
B3 / 39	Metal continuity and the removal term	Derivation, well-mixed removal assumption and linear abundance convention
B4 / 40	Selection and uncertainty in the occupancy comparison	Cut versus uncut denominator; equal-galaxy weighting; whole-galaxy bootstrap values from the existing report
B5 / 41	What CO can and cannot measure	Molecular mass, depletion time, supply-rate degeneracy and conversion assumptions
B6 / 42	References and related MAUVE results	Paper links and source details for follow-up

Delivery and rehearsal guidance

The slides use existing scientific figures and images with source citations in speaker notes. New summary tables and charts remain editable. The notes contain slide time budgets, cumulative checkpoints, explanation cues, transitions and scientific limits.

Rehearse to the 5-, 11-, 24-, 32:30- and 40-minute checkpoints. If running late, shorten the two outflow examples by 20 seconds each, compress the one-disc spatial example by 30 seconds, and shorten the regulator-literature slide to its final contribution line. Preserve the baryon-cycle-to-environment bridge, the diagnostic caveat, the scale-free analytic mapping, the occupancy definitions and the NGC4254 component distinction. This offers approximately two minutes of recovery without changing the scientific argument.

The stage results are an existing September analysis snapshot and have not been recomputed for this talk. The new work uses those stored figures and CSV values. Paper I and the survey results are sourced from their papers. The NGC4254 metallicity-fluctuation figure is a user-provided preliminary snapshot and was not regenerated. The talk does not claim a newly executed science pipeline, a measured local gas-supply-rate map, or a sample-wide spiral-arm result.

Sources and figure provenance

All local paths below are relative to /Users/Igniz/Desktop/ICRAR/.

S0 - Talk brief: [Seminar Package Advice](#), especially the latest user revision of the Yonsei scope and title. The quoted invitation specifies the 45-minute talk and 15-minute discussion on 22 September.

S1 - Survey: Catinella et al. (2025), *Multiphase Astrophysics to Unveil the Virgo Environment (MAUVE)*, The Messenger 195, 15-18. [DOI](#). Local: EAS2026/pdfs/Catinella2025.pdf; survey material also in EAS2026/EAS2026 SS7 Rongjun Huang.pptx.

S2 - Ionized outflow: Watts et al. (2024), *MAUVE: a 6 kpc bipolar outflow launched from NGC4383, one of the most HI-rich galaxies in the Virgo cluster*. [Paper](#). Local: EAS2026/pdfs/Watts2024.pdf.

S3 - Cold outflow: Cortese et al. (2026), *MAUVE: Cold neutral gas in the outflow of NGC4383 and evidence for a fountain flow*. [Author version](#). Local: EAS2026/pdfs/Cortese2026.pdf and EAS2026/panels/Cortese2026.png.

S4 - Stripped-galaxy feedback: Attwater et al. (2026), *MAUVE-MUSE: A Star Formation-driven Outflow Caught in the Act of Quenching the Stripped Virgo Galaxy NGC4064*. [Author version](#). Local: EAS2026/pdfs/Attwater2026.pdf and EAS2026/panels/Attwater2026.png.

S5 - Ionization and kinematics: Brown et al. (2026), *MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks*. [Author version](#). Local: EAS2026/pdfs/Brown2026.pdf and EAS2026/panels/Brown2026.png.

S6 - Main result and model: Huang et al. (2026), *MAUVE-MUSE: When Metallicity Follows or Fights Star Formation - A Mass-Dependent Inversion in Virgo Galaxies*. [Author version](#). Local: EAS2026/pdfs/Huang2026.pdf; manuscript MAUVE/MAUVE Rongjun Zgas-Sigma_SFR paper/main.tex. Final figure crops come directly from the local paper PDF, retaining its figure versions and numbering. Older standalone figures in extended/Plot_Paper_1/ were inspected but are not substituted for the paper figures.

S7 - Existing talk material: EAS2026/EAS2026 SS7 Rongjun Huang.pptx and EAS2026/Group MEETING Rongjun Huang.pptx. Existing reference images and reservoir illustration are reused with their provenance. The paper's correct dispersion selection overrides the old EAS typo.

S8 - Occupancy analysis snapshot: further/20260905_stage_SF_fraction_analysis.md and further/20260905_stage_SF_fraction_analysis_assets/. Numeric charts use stage_global_equal_uncut.csv and stage_global_equal_snr25.csv; other evidence uses the existing radial-profile, individual-map and bootstrap products. No science pipeline rerun is implied.

S9 - Latest scientific framing: MAUVE/20260906 Where Star Formation Remains along Infall Stages.md, with its referenced existing figures. Where its shorthand interpretation is stronger than the diagnostic evidence, the talk uses the explicit qualifications from S8.

S10 - Reservoir definitions:

MAUVE/20260831_Molecular_Gas_Supply_Definitions_and_Observational_Inference.md. Used to distinguish molecular mass, replenishment, transport and chemical conversion rates.

S11 - Baryon-cycle artwork: conceptual baryon-cycle image embedded in 202609Korea/20260922_Yonsei_CGER.pptx, slide 2. It is used only to orient the audience to inflow, star formation, feedback and recycling; no quantitative inference is taken from the artwork.

S12 - Original MAUVE science justification: MAUVE/MAUVE_sci_just_compressed.pdf, especially the scientific rationale on pages 2-4 and Figure 1 on page 5. This source motivates the surviving-inner-disc question, the competing mechanisms for altered H I/H2 and star formation, and the need to resolve the entire molecular disc across infall stages.

S13 - PHANGS resolved-disc benchmark: Emsellem et al. (2022), *The PHANGS-MUSE survey: Probing the chemo-dynamical evolution of disc galaxies*, A&A 659, A191 ([DOI](#)); Leroy et al. (2021), *PHANGS-ALMA: Arcsecond CO(2-1) Imaging of Nearby Star-Forming Galaxies*, ApJS 257, 43 ([DOI](#)). The displayed local image is `v3tk_to_VRI/NGC4254_observed_VRI.png`. PHANGS is used as the resolved nearby-disc benchmark, not described as a purely isolated sample; NGC4254 is a Virgo member.

S14 - Regulator-model precedents: Forbes et al. (2014), *On the origin of the fundamental metallicity relation and the scatter in galaxy scaling relations*, MNRAS 443, 168-185 ([DOI](#)); Wang and Lilly (2021), *Gas-phase Metallicity as a Diagnostic of the Drivers of Star Formation on Different Spatial Scales*, ApJ 910, 137 ([DOI](#)). Forbes et al. foreground stochastic accretion and discuss parameter-scatter alternatives. Wang and Lilly derive approximate small-amplitude responses for varying inflow and SFE and verify the response numerically, including large-amplitude tests.

S15 - Preliminary NGC4254 metallicity fluctuations: user-provided four-panel snapshot supplied on 10 September 2026 and copied into the presentation build as `NGC4254_metallicity_fluctuations_preliminary.jpg`. The meanings of $C_{\{2p,0\}}$, $C_{\{2p,N\}}$, C_M and C_{NP} follow Rongjun Huang's supplied interpretation. The figure is presented as preliminary one-galaxy evidence; no notebook or science pipeline was rerun for the talk.

S16 - Luca Cortese MAUVE overview: `202609Korea/cortese_ascona26.pdf`, *MAUVE: Multiphase views of gas cycling and quenching in the Virgo cluster*, especially pages 11-20. Used for the connected early-science overview, the “feedback at all infall stages” framing, the multiphase NGC4383 sequence, the late-stage NGC4064 interpretation, and the DIG synthesis.

S17 - Toby Brown MAUVE colloquium: `202609Korea/mauve-muse_colloquium.pdf`, *How does environment affect the ISM in Virgo Cluster galaxies?*, especially pages 12-26. Used for the overview-to-evidence-chain structure and the launch / phase transition / dominant phase / fate progression. Quantitative claims were checked against the corresponding Watts, Cortese, Attwater and Brown papers rather than taken from the slides alone.

Image assets: `further/20260907_MAUVE_seminar_poster_images/`, particularly the representative-field mosaic and galaxy RGBs, with `README.md` and `provenance.json`. These are MUSE-derived colour composites; approximately 100 pc refers to effective resolution, not pixel sampling. The mosaic is cropped only for slide framing while keeping the displayed galaxy structures and their labels intact.

The regulator-literature metadata and claims were checked against the primary papers on 10 September 2026. Dates and counts in older local files are retained only when explicitly attached to their corresponding analysis snapshot.